

ENGR 102 Summer Bridge

Day 12 Student Quiz

For Loops, Range Bounds, Counters, and Accumulators

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as `and`, `or`, and `not`. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Trace every range-driven state update.

5 points

```
total = 0
count = 0

for number in range(2, 10, 2):
    total = total + number
    if total >= 10:
        count = count + 1
    print(number, total, count)
```

List range values, then give number, total, and count after every pass and all exact output.

Question 2. Repair an excluded-stop defect.

3 points

```
limit = 4
total = 0

for number in range(1, limit):
    total = total + number

print(total)
```

The intent is 1 through limit inclusive. Give actual, intended, cause, repair, and one smallest boundary test.

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Question 3. Program construction

12 points

Write one complete Python program that satisfies every requirement.

- Read `sample_count` as an integer and `threshold` as a float.
- If `sample_count` is not positive, print `invalid` and request no readings.
- Otherwise read exactly `sample_count` floating-point readings.
- Compute total, average, and the count of readings at or above `threshold`.
- Print `Total` to one decimal, `Average` to two decimals, and `At or above` as an integer on separate labeled lines.

[illegible]

Test evidence / expected result